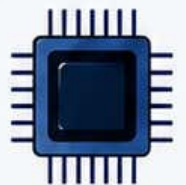
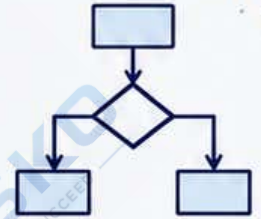


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Most Important Questions

INSTRUCTIONS



1. Understand the Concepts

Go through the basic concepts related to each question before practising.



2. Practice Regularly

Solve these important questions and their variations for better understanding.



3. Focus on Exam Pattern

These questions are selected as per previous year papers and latest syllabus.



4. Attempt Within Time Limit

Practice under timed conditions to improve speed and accuracy.



5. Revise and Self-Check

Review your answers, identify weak areas and revise regularly to score higher.

Learn • Practice • Improve • Succeed

YOUR HARD WORK PAYS OFF!



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Data Structure — Most Important Questions

Unit I - Data Structure & Array

- **Answers in one or two lines (1 mark questions)**

1. What is a data structure?
2. What is a primitive data structure?
3. What is a non-primitive data structure?
4. What is a linear data structure?
5. What is a non-linear data structure?
6. What is an array?
7. What is the memory representation of an array?
8. What is array traversal?
9. What is insertion in an array?
10. What is deletion in an array?

- **Short Answer Questions (3 marks questions)**

1. What is Data Structure ? Explain primitive and non-primitive data structures.
2. What is Data Structure ? Differentiate between linear and non-linear data structures.
3. Explain the basic operations of data structures.
4. Explain array definition, concepts and memory representation.
5. Explain array traversal.
6. Explain insertion and deletion operations in an array.

- **Long Answer Questions (6 marks questions)**

1. Explain data structure, its types and basic operations in detail.
2. Explain arrays with memory representation and basic concepts.
3. Explain array operations — Traversing, Insertion and Deletion — with examples

Unit II - Stack & Queue

- **Answers in one or two lines (1 mark questions)**

1. What is a stack?
2. What is LIFO?
3. What is stack memory representation?
4. What is stack traversal?
5. What is stack insertion?
6. What is stack deletion?
7. What is a queue?
8. What is FIFO?
9. What is queue memory representation?
10. Name the types of queues.

- **Short Answer Questions (3 marks questions)**

1. What is Stack ? Explain stack and its basic concepts.
2. What is Stack ? Explain stack memory representation.
3. What is Stack ? Explain stack operations: Traversing, PUSH and POP.
4. What is Queue ? Explain queue and its basic concepts.
5. What is Queue ? Explain queue memory representation.
6. What is Queue ? Explain queue operations: Traversing, Insertion and Deletion.
7. Explain different types of queues.

- **Long Answer Questions (6 marks questions)**

1. Explain Stack with memory representation and its operations.
2. Explain Queue with memory representation and its operations.
3. What is Queue ? Explain different types of queues.
4. Differentiate between Stack and Queue.
5. What is Stack ? Write Algorithms of PUSH & POP Operations .

Unit III - Linked List & Tree

- **Answers in one or two lines (1 mark questions)**

1. What is a linked list?
2. What is a node?
3. What is the memory representation of a linked list?
4. Name the types of linked lists.
5. What is a tree?
6. Define root node.
7. What is a leaf node?
8. What is a binary tree?
9. What is a complete binary tree?
10. What is a Binary Search Tree (BST)?
11. What is tree traversal?
12. Name the types of tree traversal.

- **Short Answer Questions (3 marks questions)**

1. What is a Linked List ? Explain a linked list and its basic concepts.
2. Explain memory representation of linked list.
3. Explain types of linked lists.
4. Explain linked list operations: Traversing, Insertion and Deletion.
5. What is a Tree ? Explain tree terminology.
6. Explain memory representation of trees.
7. Explain Binary Tree, Complete Binary Tree and Binary Search Tree.
8. Explain Preorder, Inorder and Postorder traversal.

- **Long Answer Questions (6 marks questions)**

1. Explain linked list, its memory representation, types and operations.
2. What is a Linked List ? Explain different types of linked lists with suitable diagrams.
3. What is a Tree ? Explain trees, their terminology and memory representation.
4. What is a Tree ? Explain Binary Tree, Complete Binary Tree and Binary Search Tree.
5. What is a Tree ? Explain tree traversals — Preorder, Inorder and Postorder — with examples.
6. Explain insertion and deletion operations in a tree.

Unit IV - Searching & Sorting

- **Answers in one or two lines (1 mark questions)**

1. What is searching?
2. What is sorting?
3. What is Linear Search?
4. What is Binary Search?
5. What is Indexed Sequential Search?
6. What is Bubble Sort?
7. What is Selection Sort?
8. What is Insertion Sort?

- **Short Answer Questions (3 marks questions)**

1. What is searching? Explain Linear Search with an example.
2. What is searching? Explain Binary Search with an example.
3. What is searching? Explain Indexed Sequential Search.
4. Differentiate between Linear Search and Binary Search.
5. What is sorting ? Explain Bubble Sort with an example.
6. What is sorting ? Explain Selection Sort with an example.
7. What is sorting ? Explain Insertion Sort with an example.

- **Long Answer Questions (6 marks questions)**

1. What is searching? Explain Linear Search and Binary Search with suitable examples.
2. What is searching? Explain Indexed Sequential Search in detail.
3. What is searching? Explain Bubble Sort with an algorithm and step-by-step example.
4. What is Sorting? Explain Selection Sort with an algorithm and step-by-step example.
5. What is Sorting? Explain Insertion Sort with an algorithm and step-by-step example.
6. Compare different searching and sorting techniques.